Computational Chemistry Comparison and Benchmark DataBase Release 22 (May 2022)
Standard Reference Database 101 National Institute of Standards and Technology

Experimental data for CO (Carbon monoxide)

22 02 02 11 45

Other names
Carbon monoooxide; Carbon monoxide; Carbon oxide; Carbon oxide (CO); Carbone; Carbonic oxide; Carbonio; Kohlenmonoxid; Kohlenoxyd; Koolmonoxyde; Monoxide; NA 9202; Oxyde de carbone;

INChI	INChIKey	SMILES	IUPAC name
InChI=1S/CO/c1-2	UGFAIRIUMAVXCW-UHFFFAOYSA-N	[C]#O	Carbon monoxide

State	Conformation
$^1\Sigma$	C*V

Enthalpy of formation (Hfg), Entropy, Integrated heat capacity (0 K to 298.15 K) (HH), Heat Capacity (Cp)

Property	Value	Uncertainty	units	Reference	Comment
Hfg(298.15K) $\Delta_f H°$	-110.53	0.17	kJ mol^{-1}	CODATA	
Hfg(0K) $\Delta_f H°$	-113.81	0.17	kJ mol^{-1}	CODATA	
Entropy (298.15K) $S°$	197.66	0.00	J K^{-1} mol^{-1}	CODATA	
Integrated Heat Capacity (0 to 298.15K) $H°-H°(T)$	8.67	0.00	kJ mol^{-1}	CODATA	
Heat Capacity (298.15K) $C_p°$	29.14		J K^{-1} mol^{-1}	Gurvich	

Information can also be found for this species in the NIST Chemistry Webbook

Vibrational levels (cm^{-1})

Mode Number	Symmetry	Frequency			Intensity			Comment	Description
		Fundamental(cm^{-1})	Harmonic(cm^{-1})	Reference	(km mol^{-1})	unc.	Reference		
1	Σ	2143	2170	1979HUB/HER					

Detailed diatomic data

ω_e	$\omega_e x_e$	$\omega_e y_e$	B_e	α_e	ZPE	reference
2169.756	13.28803	0.0104109	1.931602	1.750513E-02	1081.747	2007Iri:389

vibrational zero-point energy: 1071.6 cm^{-1} (from fundamental vibrations)
Calculated vibrational frequencies for CO (Carbon monoxide).

More spectroscopic constants are available at the NIST Physics Laboratory website:
http://physics.nist.gov/PhysRefData/MolSpec/Diatomic/Html/Tables/CO.html

Rotational Constants (cm^{-1})

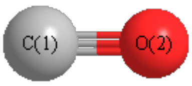
See section I.F.4 to change rotational constant units

A	B	C	reference	comment
	1.92253		NISTdiatomic	B0

Calculated rotational constants for CO (Carbon monoxide).

Product of moments of inertia $I_A I_B I_C$				
8.768466	amu Å^2		1.456061E-39	gm cm^2

Geometric Data



Point Group $C_{\infty v}$

Internal coordinates

distances (r) in Å, angles (a) in degrees, dihedrals (d) in degrees

Description	Value	unc.	Connectivity				Reference	Comment
			Atom 1	Atom 2	Atom 3	Atom 4		
rCO	1.128	0.000	1	2			NISTdiatomic	re

Cartesians

Atom	x (Å)	y (Å)	z (Å)
C1	0.0000	0.0000	0.0000
O2	0.0000	0.0000	1.1282

Atom - Atom Distances

Distances in Å

	C1	O2
C1		**1.1282**
O2	**1.1282**	

Calculated geometries for CO (Carbon monoxide).

Bond descriptions

Examples: C-C single bond, C=C, double bond, C#C triple bond, C:C aromatic bond

Bond Type	Count
C#O	1

Connectivity

Atom 1	Atom 2
C1	O2

Electronic energy levels (cm^{-1})

Energy (cm^{-1})	Degeneracy	reference	description
0	1	1979HUB/HER	$^1\Sigma$

Ionization Energies (eV)

Ionization Energy	I.E. unc.	vertical I.E.	v.I.E. unc.	reference	Comment
14.014	0.000	14.014		webbook	questionable EA of 1.3703 eV

Proton Affinity (kJ mol^{-1})

Proton Affinity	unc.	Product	reference	comment
594.0		HCO^+	webbook	HCO+
426.3		COH^+	webbook	COH+

Dipole, Quadrupole and Polarizability

Electric dipole moment

State	Config	State description	Conf description	Exp. min.	Dipole (Debye)				Reference	comment	Point Group	Components	
					x	y	z	total				dipole	quadrupole
1	1	$^1\Sigma$	$C_{\infty v}$	True				0.112	NSRDS-NBS10	MW	$C_{\infty v}$	1	1
2	1	$^3\Pi$	$C_{\infty v}$	True				1.375	1970Ste/Gam:3467-3482	Omega=2. For Omega=1 1.37454 D	$C_{\infty v}$	1	1

Experimental dipole measurement abbreviations: MW microwave; DT Dielectric with Temperature variation; DR Indirect

(usually an upper limit); MB Molecular beam
Calculated electric dipole moments for CO (Carbon monoxide).

Electric quadrupole moment

State	Config	State description	Conf description	Exp. min.	Quadrupole (D Å)			Reference	comment	Point Group	Components	
					xx	yy	zz				dipole	quadrupole
1	1	$^1\Sigma$	$C_{\infty v}$	True	1.420	1.420	-2.839	1998Gra/Imr:49	-9.47+-0.15 E-40 Cm^2	$C_{\infty v}$	1	1
2	1	$^3\Pi$	$C_{\infty v}$	True						$C_{\infty v}$	1	1

Calculated electric quadrupole moments for CO (Carbon monoxide).

Electric dipole polarizability (Å^3)

alpha	unc.	Reference
1.953		1997Oln/Can:59

Calculated electric dipole polarizability for CO (Carbon monoxide).

References

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squib	reference	DOI
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1979HUB/HER	Huber, K.P.; Herzberg, G., Molecular Spectra and Molecular Structure. IV. Constants of Diatomic Molecules, Van Nostrand Reinhold Co., 1979	10.1007/978-1-4757-0961-2
1997Oln/Can:59	TN Olney, NM Cann, G Cooper, CE Brion, Absolute scale determination for photoabsorption spectra and the calculation of molecular properties using dipole sum-rules, Chem. Phys. 223 (1997) 59-98	10.1016/S0301-0104(97)00145-6
1998Gra/Imr:49	C Graham, DA Imrie, RE Raab "Measurement of the electric quadrupole moments of CO2, CO, N2, Cl2 and BF3" Mol. Phys. 93(1), 1998, 49-56	10.1080/002689798169429
2007Iri:389	KK Irikura "Experimental Vibrational Zero-Point Energies: Diatomic Molecules" J. Phys. Chem. Ref. Data 36(2), 389, 2007	10.1063/1.2436891
CODATA	Cox, J.D.; Wagman, D.D.; Medvedev, V.A.CODATA Key Values for Thermodynamics. Hemisphere, New York, 1989	10.1002/bbpc.19900940121
Gurvich	Gurvich, L.V.; Veyts, I. V.; Alcock, C. B., Thermodynamic Properties of Individual Substances, Fouth Edition, Hemisphere Pub. Co., New York, 1989	
NISTdiatomic	NIST Diatomic Spectral Database (www.physics.nist.gov/PhysRefData/MolSpec/Diatomic/index.html)	10.18434/T4T59X
NSRDS-NBS10	R. D. Nelson Jr., D. R. Lide, A. A. Maryott "Selected Values of electric dipole moments for molecules in the gas phase" NSRDS-NBS10, 1967	10.6028/NBS.NSRDS.10
webbook	NIST Chemistry Webbook (http://webbook.nist.gov/chemistry)	10.18434/T4D303

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BO+ (boron monoxide cation)	CO- (carbon monoxide anion)